▲ LEG 5 · AUTOMOTIVE AI · DAYS 31–37

THE AI ROAD TRIP LEG 5

7 new self-driving AI papers.
Rebuilt from scratch in 7 days.

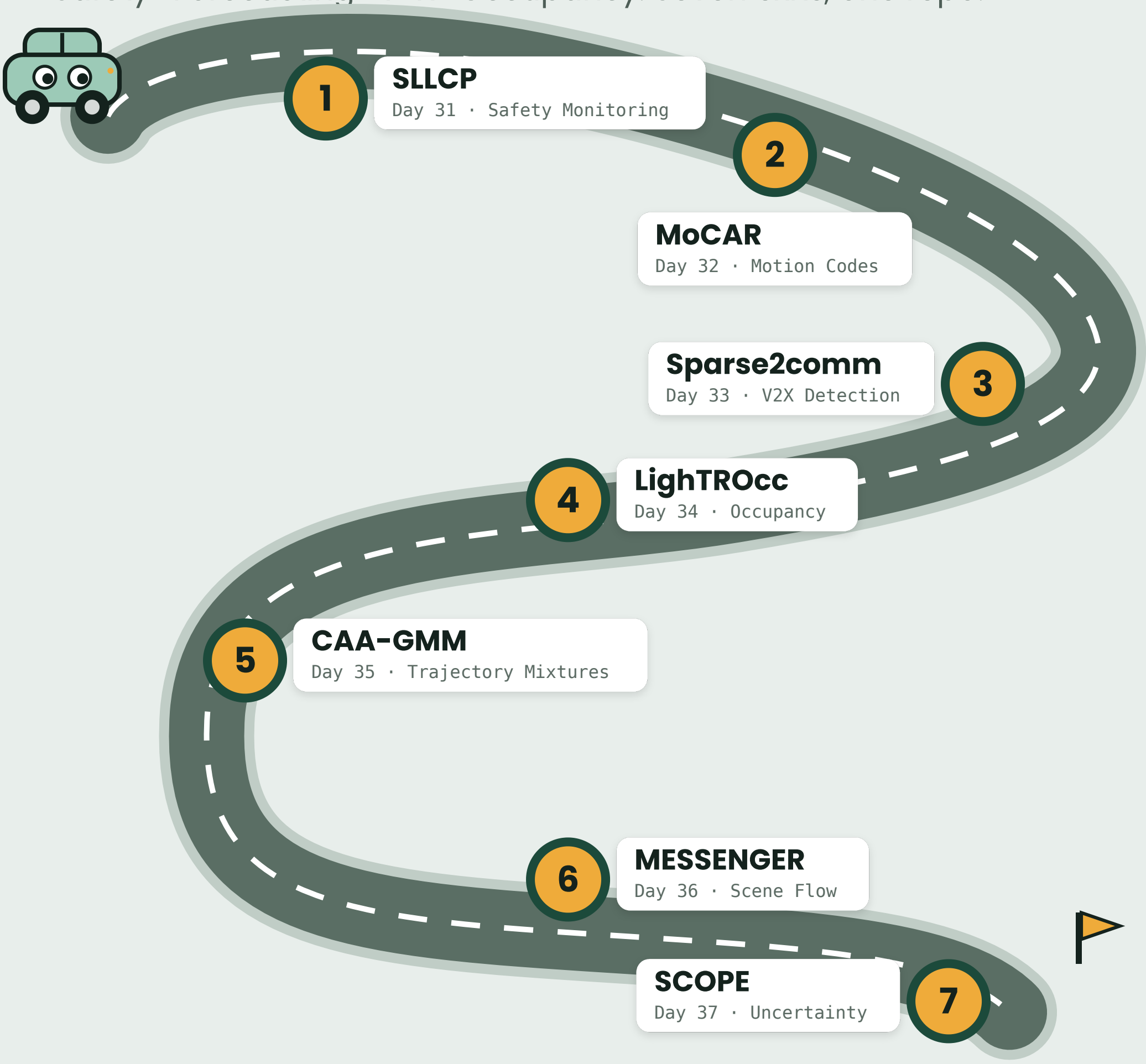
D31 D32 D33 D34 D35 D36 D37



Swipe to take Exit 01 →

This week's route

Safety · Forecasting · V2X · Occupancy. Seven exits, one repo.



7	7	7	37
PAPERS	PYTORCH REBUILDS	SIMULATION GIFS	SYSTEMS SO FAR

Swipe for Exit 01 →

01

EXIT 01 · SAFETY MONITORING

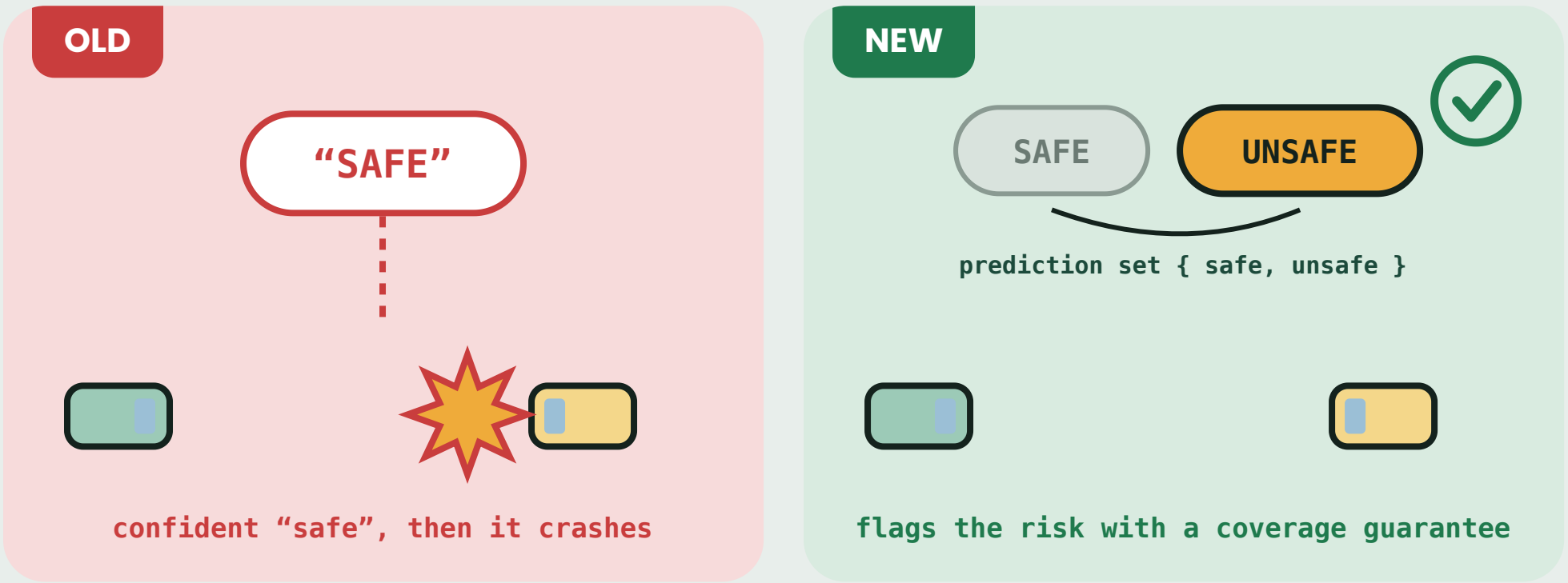
SLLCP · arXiv 2610.02765

L. Marques, R. Fang … D. Berenson · Univ. of Michigan

Makes an AI safety guard admit when it isn't sure

A vision-language model judging a driving plan loves to say “safe”. This layer wraps its scores in conformal prediction, so unsafe plans get flagged with a coverage guarantee.

> *Like a smoke alarm tuned for the rare fire, not the average day.*



0.1 → 89.5% unsafe-plan recall in my rebuild (paper: 4.6 → 89.6%)

✓ REPRODUCED IN MY REBUILD

02

EXIT 02 · TRAJECTORY FORECASTING

MoCAR · arXiv 2610.06210

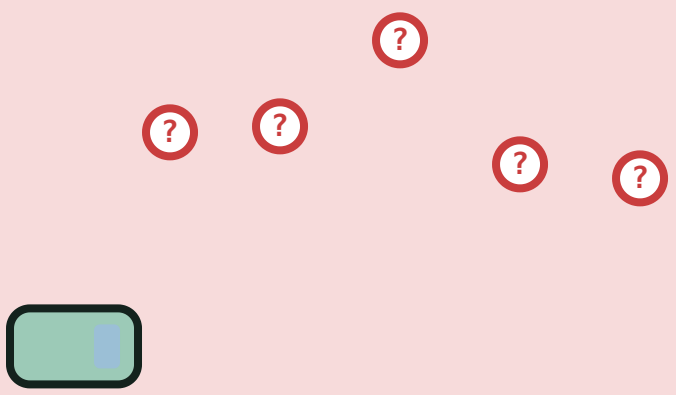
Authors not recovered (abstract-level read)

Writes a car's future one motion code at a time

Instead of predicting raw x-y points, a language-model-style decoder predicts learned motion codes, each carrying the change of reference frame it causes.

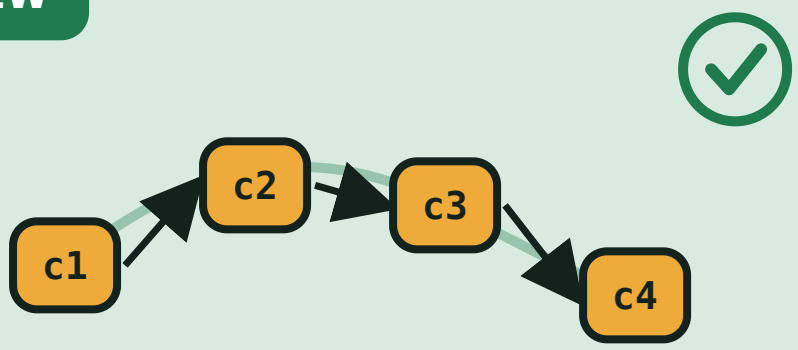
> *Like dictating a route turn by turn, where each turn changes where you stand.*

OLD



decodes every point in one shot

NEW



each code shifts the reference frame

0.133 vs 0.176 minADE@3 in m: simple one-shot beat MoCAR in my rebuild

× NOT AT TOY SCALE

03

EXIT 03 · V2X · COOPERATIVE DETECTION

Sparse2comm · arXiv 2610.08573

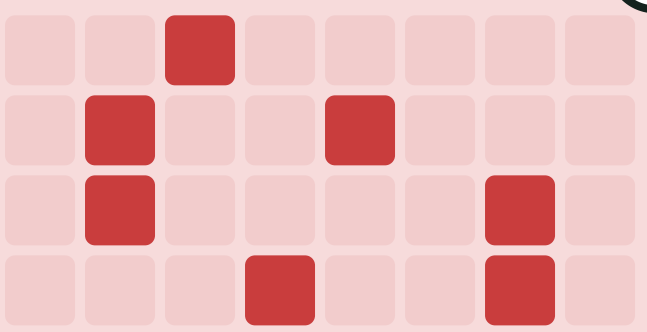
L. Yang, B. Li, C. Lin … C. Lv

Sees through a bad V2X link using ~2% of the data

A neighbour sends a sliver of features. The ego rebuilds the rest, rewinds the delay with predicted motion flow, and corrects spatial offset before fusing.

> *Like finishing a crossword from a smudged fax, then checking the clock and the map.*

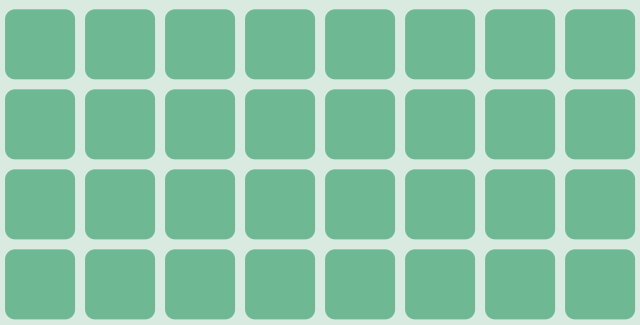
OLD



holes · stale · skewed

fuses the broken message as-is

NEW



REBUILD REWIND REALIGN

≈2% of the data, repaired in order

0.814 → 0.865 AP vs my own sparse-fusion baseline (paper: +12–20 AP vs Where2comm)

✓ REPRODUCED IN MY REBUILD

04

EXIT 04 · OCCUPANCY FORECASTING

LighTROcc · arXiv 2610.09444

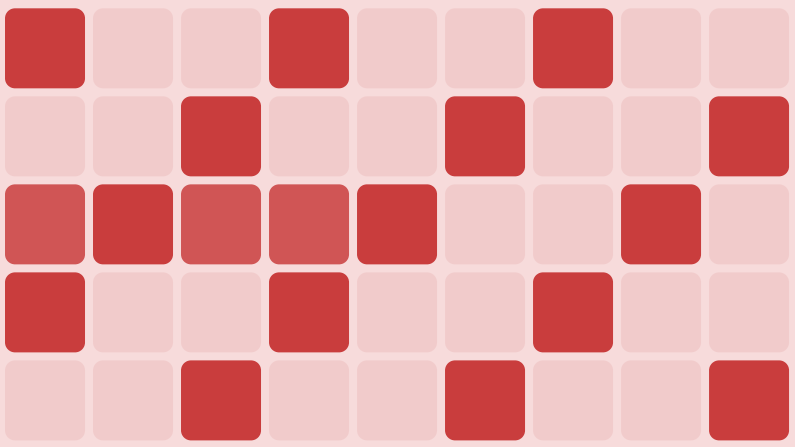
H. Jung, S. Kim … S. Kim · Korea Univ · UT Austin · KAIST

Forecasts free space as a few moving Gaussians, not a giant grid

One query per movable object is lifted to 3D by depth, expanded into Gaussians, and slid forward in time: future occupancy is the same object, moved.

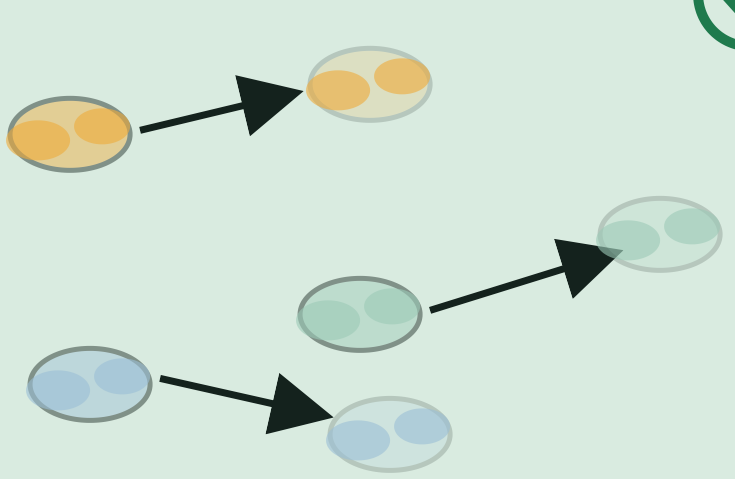
> *Like tracking three balloons instead of repainting the whole sky.*

OLD



dense voxel grid: cost grows with space
blobs have no identity

NEW



a few Gaussians per object
identity kept; moved forward

0.40 vs 0.24 IoU at equal steps. At equal wall-clock a dense grid ties (0.43)

◐ PARTLY REPRODUCED

05

EXIT 05 · TRAJECTORY FORECASTING

CAA-GMM · arXiv 2610.09174

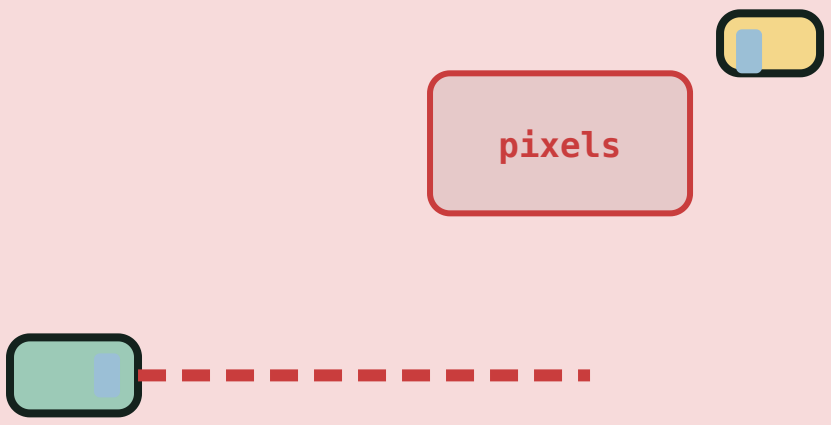
A. Raftari, B. Ebrahimi Soorchaei, Y. Fallah · UCF · AMD

Decides who and what to listen to, until the radio cuts out

Attention weighs neighbours against map cues and outputs a Gaussian mixture, one interpretable mode per future. The catch: what if V2X messages vanish?

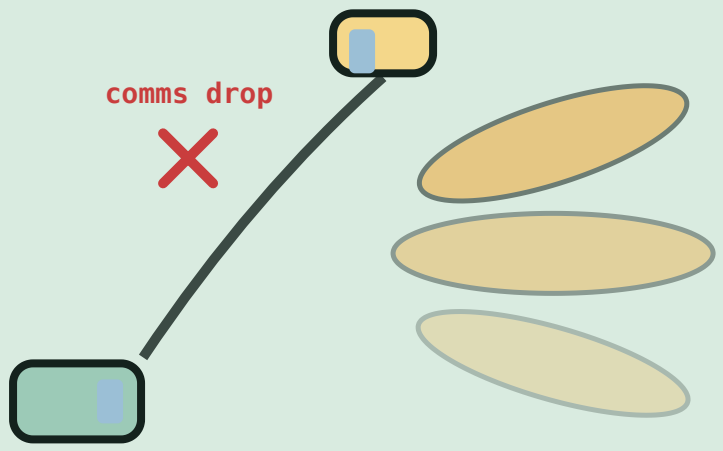
> *Like a driver who trusts the car ahead, until its brake lights go dark.*

OLD



neighbours are blobs in a raster
one path, no stated uncertainty

NEW



attention + K Gaussian modes
uncertainty per mode

0.055 vs 0.136 minADE@6 in m, clean comms. Total outage: 0.578 vs 0.213 (worse)

◐ PARTLY REPRODUCED

06

EXIT 06 · LIDAR SCENE FLOW

MESSENGER · arXiv 2610.10759

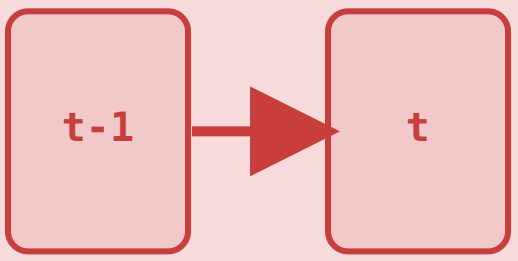
J. Liu, J. Li … P. O. Kristensson · Cambridge · SJTU

Remembers how each point was moving to forecast where it goes

Instead of comparing two LiDAR frames, a memory of past flows and latent states initialises the next forecast. An uncertainty gate should filter bad memories.

> *Like predicting a thrown ball from its whole arc, not two snapshots.*

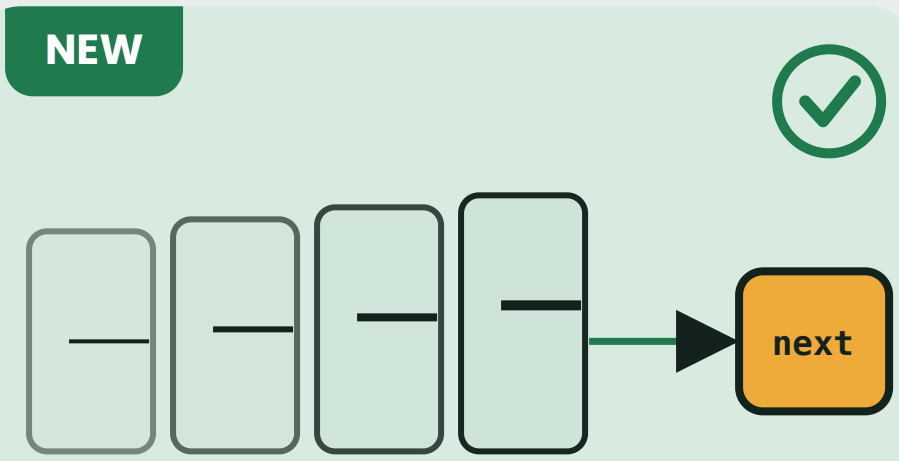
OLD



sees two frames only

extrapolates blind

NEW



memory of past flows + uncertainty gate

starts from a trend, not a snapshot

-8.5% error at +2 s in my rebuild (paper: -71.6% / -67.7%). Gate didn't learn

◐ PARTLY REPRODUCED

07

EXIT 07 · UNCERTAINTY · TRAJECTORY DIFFUSION

SCOPE · arXiv 2610.12431

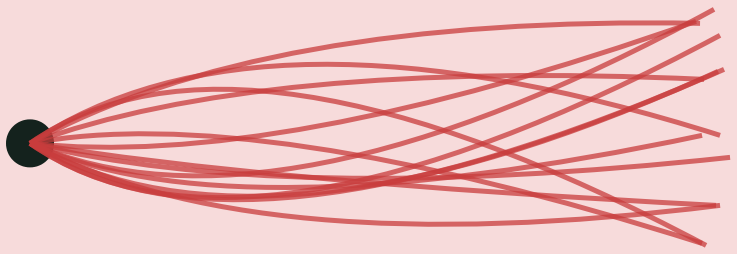
Z. Xue, J. Kam … V. Bonnet, H. Soh · CoRL 2026 Spotlight

Draws a calibrated uncertainty tube in 0.7 ms instead of 30

A small head distilled from a diffusion model's own score curvature predicts per-step Gaussian tubes around a nominal path, with no repeated sampling.

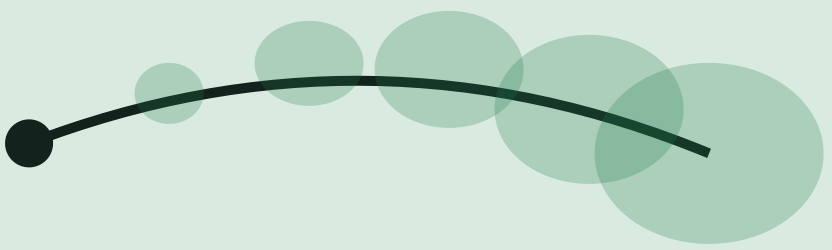
> *Like drawing a chalk outline once instead of tracing 32 shaky copies.*

OLD



32 diffusion samples → covariance
~30 ms per scene

NEW



one head forward, one small solve
~0.7 ms per scene

45× faster than 32-sample Monte-Carlo, with 95% tube coverage in every crowd bucket

✓ REPRODUCED IN MY REBUILD

REALITY CHECK

The honest scorecard

I rebuild every paper from scratch in PyTorch on synthetic data. Here's what held up, and what didn't.

3	3	1
HELD UP	PARTLY	NOT AT TOY SCALE

01	SLLCP	Paper: unsafe detection 4.6→89.6% (Qwen) **My rebuild:** recall 0.1→89.5%; localization only +0.8 pt	✓ Held up
02	MoCAR	Paper: top-tier Argoverse (no numbers recovered) **My rebuild:** one-shot 0.133 m beats MoCAR 0.176 m	× Not at toy scale
03	Sparse2comm	Paper: +12–20 AP@0.5 vs Where2comm **My rebuild:** AP 0.814→0.865 vs my own baseline	✓ Held up
04	LighTROcc	Paper: beats dense baselines (no tables) **My rebuild:** wins per step, ties dense per wall-clock	◐ Partly
05	CAA-GMM	Paper: beats raster SOTA, robust to comm loss **My rebuild:** 0.055 vs 0.136 m, but fails total outage	◐ Partly
06	MESSENGER	Paper: EPE3D -71.6% / -67.7% **My rebuild:** -8.5% at +2 s; worse on static points	◐ Partly
07	SCOPE	Paper: Maze2D 99.25% vs 96.35% success **My rebuild:** 95% tube coverage everywhere, 45× faster	✓ Held up

A toy-scale miss isn't a verdict on the paper. It's a signal about what the idea needs: scale, real data, or details the abstract doesn't show.

37 SYSTEMS · 37 DAYS

Want the code?

Tested PyTorch implementations, simulation GIFs and honest write-ups for all 37 systems so far. Free and open on GitHub.

```
github.com/Lalithprabu/
automotive-ai-daily-reviews
```

★ Star the repo

Follow for Leg 6 →

